According to part IV Article 51A (h) of Indian Constitution *"It shall be the duty of every citizen of India to develop scientific temper, humanism and spirit of inquiry and reform"*. This is the aim of our Science Education.

1.1.1 The Scientific Method

The scientific method is a step-by-step approach in studying natural phenomena and establishing laws which govern these phenomena. Any scientific method involves the following general features.

(i) Systematic observation
(ii) Controlled experimentation
(iii) Qualitative and quantitative reasoning
(iv) Mathematical modeling
(v) Prediction and verification or falsification of theories

Example

Consider a metalic rod being heated. When one end of the rod is heated, heat is felt at the other end. The following questions can be asked on this observation

a) What happens within the rod when it is heated?
b) How does the heat reach the other end?
c) Is this effect true for all materials?
d) If heat flows through the material, is it possible to visualize heat?

The process of finding the answers to these queries is scientific investigation.

The basic phenomenon of heat is discussed in unit 8.



The name Physics was introduced by Aristotle in the year 350 BC

1.2 PHYSICS - INTRODUCTION

The word 'physics' is derived from the Greek word ***"Fusis"***, meaning nature. The study of nature and natural phenomena is dealt within physics. Hence physics is considered as the most basic of all sciences.

Unification and ***Reductionism*** are the two approaches in studying physics. Attempting to explain diverse physical phenomena with a few concepts and laws is unification. For example, Newton's universal law of gravitation (in unit 6) explains the motion of freely falling bodies towards the Earth, motion of planets around the Sun, motion of the Moon around the Earth, thus unifying the fundamental forces of nature.

An attempt to explain a macroscopic system in terms of its microscopic constituents is ***reductionism***. For example, thermodynamics (unit 8) was developed to explain macroscopic properties like temperature, entropy, etc., of bulk systems. The above properties have been interpreted in terms of the molecular constituents (microscopic) of the bulk system by kinetic theory (unit 9) and statistical mechanics.

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